MAT 203 :Discrete Mathematics

Module 1: Fundamentals of Logic

Tony Augustine

Assistant Professor

Department of Science & Humanities

Viswajyothi college of Engineering & Technology, Vazhakulam

Outline

- **Mathematical logic - Basic connectives and truth table.**
- **Statements, Logical Connectives.**
- **Tautology, Contradiction.**
- **Logical Equivalence - The Laws of Logic.**
- **The Principle of duality.**
- **Substitution Rules.**
- **The implication - The Contrapositive, The Converse, The Inverse.**
- **Logical Implication - Rules of Inference.**
- **The use of Quantifiers - Open Statement, Quantifier.**
- **Logically Equivalent – Contrapositive, Converse , Inverse.**
- **Logical equivalences and implications for quantified statement.**
- **Implications , Negation.**

Logic

- Mathematical logic is the discipline that deals with reasoning.
- One of the aims of logic is to provide rules by which we can determine whether a particular reasoning or argument is valid.
- Important for program design.
- Used for designing electronic circuit.
- Logical reasoning is used in many disciplines to establish valid results.

Propositions

- A proposition is a statement or sentence that can be determined to be either true or false (but not both).
- We say that the truth value of a proposition is either true T (1) or false F (0).
- Sentences which are exclamatory, interrogative or imperative in nature are not propositions. Lower case letters such as p, q, r … are used to denote propositions.

Examples

Examples: $2 + 2 = 4$ True

$3 \times 3 = 8$ False

787009911 is a prime

Non-examples: $x+y>0$

$x^2+y^2=z^2$

They are true for some values of x and y
but are false for some other values of x and y.

Examples

- The following are propositions
  - Today is Monday *M*
  - The grass is wet *W*
  - It is raining *R*
- The following are not propositions
  - C++ is the best language *Opinion*
  - When is the pretest? *Interrogative*
  - Do your homework *Imperative*

Examples

"Elephants are bigger than mice."

- Is this a statement?
- Is this a proposition?
- What is the truth value of the proposition?

"$520 < 111$"

- Is this a statement?
- Is this a proposition?
- What is the truth value of the proposition?

Examples

"x < y if and only if y > x."

- Is this a statement?

Yes

- Is this a proposition?

Yes

….. because its truth value does not depend on specific values of x and y.

- What is the truth value of the proposition?

True

Primitive Statements & Compound Statements

- A proposition is said to be primitive if it cannot be broken into simpler propositions. They are statements without connectives.

Eg: Vazhakulam is in EKM district.

$2+3=5$

- A compound statement is the combination of 2 or more primitive statements. They are statements with connectives.

Eg: Roses are red, and Violets are blue.

Logical Connectives

The word or phrases used to connect two or more primitive statements are called connectives

- Most common connectives:
  - Conjunction (AND) $\wedge$
  - Disjunction (OR) $\vee$
  - Negation (NOT) $\neg$, ~
  - Exclusive (XOR) $\underline{\vee}$, $\oplus$
  - Implication (If-then) $\rightarrow$
  - Bi-Condition(If and only if) $\leftrightarrow$

Logical Connectives

(a) conjunction (AND): $p \wedge q$

(b) disjunction(inclusive OR): $p \vee q$

(c) exclusive or: $p \oplus q$

(d) implication: $p \rightarrow q$ (if p then q)

(e) biconditional: $p \leftrightarrow q$ (p if and only if q, or p iff q)

Example

p = “it is hot” q = “it is sunny”

It is hot and sunny	$p \wedge q$
It is not hot but sunny	$\neg p \wedge q$
It is neither hot nor sunny	$\neg p \wedge \neg q$

Truth Table

- A table giving truth values of a compound statement in terms of its component part is called the truth table.

Note: If there are "n" distinct components in a statement's formula, we have 2^n possible combinations of the truth values in the truth table.

Truth table of Conjunction(AND)

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

- $p \wedge q$ is true only when both p and q are true.

Example

- Let p = “A decade is 10 years”
- Let q = “A millennium is 100 years”
- $p \wedge q$ = “A decade is 10 years” and “A millennium is 100 years”
- If p is true and q is false, then conjunction is false

Truth table of Disjunction (OR)

p	q	$p \lor q$
T	T	T
T	F	T
F	T	T
F	F	F

- $p \lor q$ is false only when both p and q are false

Eg: p = "John is a programmer", q = "Mary is a lawyer"
p v q = "John is a programmer or Mary is a lawyer"

Truth table of Negation (NOT)

- Negation of P: in symbols $\neg$ p

p	$\neg$ p
T	F
F	T

- $\neg$ p is false when p is true, $\neg$ p is true when p is false
  - Eg: p : "John is a programmer"

$\neg$ p = "John is not a programmer"

Truth Table Exclusive disjunction

- "Either p or q" (but not both), in symbols $p \veebar q$

p	q	$p \veebar q$
T	T	F
T	F	T
F	T	T
F	F	F

- $p \veebar q$ is true only when p is true and q is false, or p is false, and q is true.
  - Eg: p = "John is programmer, q = "Mary is a lawyer"

    $p \veebar q$ = "Either John is a programmer or Mary is a lawyer"

Implication

- An *implication* proposition is of the form

  "If p then q"
- In symbols: $p \to q$
- Example:
  - p = " A bottle contains acid"
  - q = "A bottle has a label"
  - $p \to q$ = "If a bottle contains acid then it has a label "

Conditional propositions

- The implication of $p \rightarrow q$ can be also read as
  - If p then q
  - p implies q
  - If p, q
  - p only if q
  - q if p
  - q when p
  - q whenever p
  - q follows from p
  - p is a sufficient condition for q (p is sufficient for q)
  - q is a necessary condition for p (q is necessary for p)

Truth table of implication($p \rightarrow q$)

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

- $p \rightarrow q$ is true when both p and q are true or when p is false

Example

- If the mathematics department gets an additional 40,000 then it will hire one new faculty member.
- Let p: The Mathematics Department gets an additional 40,000 and q: The mathematics Department will hire one new faculty member.

❖ In a conditional proposition $p \rightarrow q$,

p is called the hypothesis

q is called the conclusion

Bi-Conditional($p \leftrightarrow q$)

- **Definition:** The biconditional $p \leftrightarrow q$ is the proposition that is true when p and q have the same truth values. It is false otherwise.
- Note that it is equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$

Bi-Conditional($p \leftrightarrow q$)

- The biconditional $p \leftrightarrow q$ can be equivalently read as
  - *p* if and only if *q*
  - *p* is a necessary and sufficient condition for *q*
  - if *p* then *q*, and conversely
  - *p* iff *q* (Note typo in textbook, page 9, line 3)
- Examples
  - $x>0$ if and only if x^2 is positive
  - The alarm goes off iff a burglar breaks in
  - You may have pudding iff you eat your meat

Truth table of Bi-Conditional($p \leftrightarrow q$)

- The *double implication* "p if and only if q" is defined in symbols as $p \leftrightarrow q$

p	**q**	$\mathbf{p \leftrightarrow q}$
T	T	T
T	F	F
F	T	F
F	F	T

- $p \leftrightarrow q$ is true when both p & q have same truth values.

More compound statements

- Let p, q, r be simple statements
- We can form other compound statements, such as
  - $(p \vee q) \wedge r$
  - $p \vee (q \wedge r)$
  - $(\sim p) \vee (\sim q)$
  - $(p \vee q) \wedge (\sim r)$
  - and many others…

Example: truth table of $(p\vee q)\wedge r$

p	q	r	$(p \vee q) \wedge r$
T	T	T	T
T	T	F	F
T	F	T	T
T	F	F	F
F	T	T	T
F	T	F	F
F	F	T	F
F	F	F	F

Example

- Construct the truth table for the following compound proposition

$$((p \wedge q) \vee \neg q)$$

Truth table

p	q	$p \wedge q$	$\neg q$	$((p \wedge q) \vee \neg q)$
0	0	0	1	1
0	1	0	0	0
1	0	0	1	1
1	1	1	0	1

Order of Precedence

- The negation operator has precedence over all other operators

$$\neg p \vee q = (\neg p) \vee q$$

- The conjunction operator has precedence over disjunction operator.

$$p \wedge q \vee r = (p \wedge q) \vee r$$

- Implication and biconditional propositions have lower precedence than the other operators. Among them implications has precedence than biconditional propositions.

$$p \leftrightarrow q \rightarrow r = p \leftrightarrow (q \rightarrow r)$$

Order of Precedence

- To avoid unnecessary parenthesis, the following precedences hold:
  1. Negation ($\neg$)
  2. Conjunction ($\wedge$)
  3. Disjunction ($\vee$)
  4. Implication ($\rightarrow$)
  5. Biconditional ($\leftrightarrow$)

Definitions

Definition 7. (Tautology)
A proposition is said to be a *tautology* if its truth value is T for any assignment of truth values to its components.

Example: The proposition $p \vee \neg p$ is a tautology.

Definition 8. (Contradiction)
A proposition is said to be a *contradiction* if its truth value is F for any assignment of truth values to its components.

Example: The proposition $p \wedge \neg p$ is a contradiction.

Definition 9.(Contingency)
A proposition that is neither a tautology nor a contradiction is called a *contingency*.

Tautology

- A compound proposition is a *tautology* if its truth table contains only true values for every case or contains T only in last column and is denoted by T_0
- Example: $p \rightarrow p \vee q$

p	**q**	$\mathbf{p \rightarrow p \vee q}$
T	T	T
T	F	T
F	T	T
F	F	T

Contradiction

- A compound proposition is a *contradiction* if its truth table contains only false values for every case or contains F only in last column and denoted by F_0
- Example: p ^ ~p

p	p ^ (~p)
T	F
F	F

Logical equivalence

- Two propositions are said to be *logically equivalent* if their truth tables are identical.

p	**q**	**$\sim p \vee q$**	**$p \rightarrow q$**
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

- Example: $\sim p \vee q$ is *logically equivalent* to $p \rightarrow q$

Example

Ex. Prove that $p \vee q \Leftrightarrow \neg(\neg p \wedge \neg q)$.

p	q	$p \vee q$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$	$\neg(\neg p \wedge \neg q)$
F	F	F	T	T	T	F
F	T	T	T	F	F	T
T	F	T	F	T	F	T
T	T	T	F	F	F	T

Example

$p \leftrightarrow q$ is logically equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$

p	**q**	$\mathbf{p \leftrightarrow q}$	$\mathbf{(p \rightarrow q) \wedge (q \rightarrow p)}$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	T	T

Definitions

Definition 11. (Converse)

The *converse* of a conditional proposition $p \to q$ is the proposition $q \to p$

Definition 12. (Inverse)

The *inverse* of a conditional proposition $p \to q$ is the proposition $\neg p \to \neg q$

Definition 13. (Contrapositive)

The *contrapositive* of a conditional proposition $p \to q$ is the proposition $\neg q \to \neg p$.

Variations on a proposition

- Given a proposition $p \rightarrow q$, there are other propositions that can be stated.
  - Example: If a function is not continuous, it is not differentiable.
- Contrapositive: $\neg q \rightarrow \neg p$
  - Example: If a function is differentiable, then it is continuous.
- Converse: $q \rightarrow p$
  - Example: If a function is not differentiable, then it is not continuous.
- Inverse: $\neg p \rightarrow \neg q$
  - Example: If a function is continuous, then it is differentiable.

Converse

- The *converse* of $p \rightarrow q$ is $q \rightarrow p$

p	**q**	**$p \rightarrow q$**	**$q \rightarrow p$**
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

These two propositions

are <u>not</u> logically equivalent

Contrapositive

- The *contrapositive* of the proposition $p \rightarrow q$ is $\sim q \rightarrow \sim p$.

p	**q**	**$p \rightarrow q$**	**$\sim q \rightarrow \sim p$**
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

They are logically equivalent.

Example

Let us consider the statement,

"The crops will be destroyed, if there is a flood."

Let F : there is a flood & C : The crops will be destroyed

The symbolic form is, $F \rightarrow C$.

Converse $(C \rightarrow F)$

i.e., "if the crops will be destroyed then there is flood."

Inverse $(\neg F \rightarrow \neg C)$

i.e.,"if there is no flood then the crops won't be destroyed, ."

Contrapositive $(\neg C \rightarrow \neg F)$

i.e., "if the crops won't be destroyed then there is no flood."

Problem

4. Let p, q, r, s denote the following statements:

p: I finish writing my computer program before lunch.
q: I shall play tennis in the afternoon.
r: The sun is shining.
s: The humidity is low.

Write the following in symbolic form.

a) If the sun is shining, I shall play tennis this afternoon.

b) Finishing the writing of my computer program before lunch is necessary for my playing tennis this afternoon.

c) Low humidity and sunshine are sufficient for me to play tennis this afternoon.

Ans:

1. $r \rightarrow q$
2. $q \rightarrow p$
3. $(r \vee s) \rightarrow q$

Translating English sentence into a logical expression

- "You can access the internet from campus only if you are a computer science major or you are not a freshman"

a: "You can access the internet from campus"

b: "you are a computer science major"

c: "you are a freshman"

Where a,b,c are propositional variables

- $a \rightarrow (b \vee \neg c)$

Construct the truth table for the following statements:

(i) $(p \to q) \longleftrightarrow (\neg p \vee q)$

(ii) $p \wedge (p \vee q)$

(iii) $(p \to q) \to p$

(iv) $\neg (p \wedge q) \longleftrightarrow (\neg p \vee \neg q)$

(v) $(p \vee \neg q) \to q$

(i) Let $S = (p \rightarrow q) \longleftrightarrow (\neg p \vee q)$

p	q	$\neg p$	$p \rightarrow q$	$\neg p \vee q$	S
T	T	F	T	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

(ii) Let $S = p \wedge (p \vee q)$

p	q	$p \vee q$	S
T	T	T	T
T	F	T	T
F	T	T	F
F	F	F	F

(iii) Let $S = (p \rightarrow q) \rightarrow p$

p	q	$p \rightarrow q$	S
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	F

(iv) Let $S = \neg(p \wedge q) \longleftrightarrow \neg p \vee \neg q$

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$	S
T	T	T	F	F	F	F	T
T	F	F	T	F	T	T	T
F	T	F	T	T	F	T	T
F	F	F	T	T	T	T	T

(v) Let $S = (p \vee \neg q) \to q$

p	q	$\neg q$	$p \vee \neg q$	S
T	T	F	T	T
T	F	T	T	F
F	T	F	F	T
F	F	T	T	F

Problem

5. Let p, q, r denote the following statements about a particular triangle ABC.

p: Triangle ABC is isosceles.
q: Triangle ABC is equilateral.
r: Triangle ABC is equiangular.

Translate each of the following into an English sentence.

a) $q \rightarrow p$

b) $\neg p \rightarrow \neg q$

c) $q \leftrightarrow r$

d) $p \wedge \neg q$

e) $r \rightarrow p$

Ans:

$q \rightarrow p$: If the Δle ABC is equilateral then Δle ABC is isosceles

$\neg p \rightarrow \neg q$: If the Δle ABC is not isosceles then it is not equilateral.

$q \leftrightarrow r$: Δle ABC is equilateral iff it is equiangular.

$p \wedge \neg q$: Δle ABC is isosceles and it is not equilateral.

$r \rightarrow p$: If Δle ABC is equiangular then it is isosceles.

Problem

Example 1.1 Construct a truth table for each of the following compound propositions:

(a) $(p \vee q) \to (p \wedge q)$;

(b) $(p \to q) \to (q \to p)$;

(c) $(q \to \neg p) \leftrightarrow (p \leftrightarrow q)$;

(d) $(p \leftrightarrow q) \leftrightarrow ((p \wedge q) \vee (\neg p \wedge \neg q))$;

(e) $(\neg p \leftrightarrow \neg q) \leftrightarrow (p \leftrightarrow q)$.

Example 1.2 Construct the truth table for each of the compound propositions given as follows:

(a) $((p \to (q \to r)) \to ((p \to q) \to (p \to r))$

(b) $\neg(p \vee (q \wedge r)) \leftrightarrow ((p \vee q) \wedge (p \to r))$

(c) $(\neg p \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$

(d) $(p \to (q \to s)) \wedge (\neg r \vee p) \wedge q$

(e) $((p \to q) \to r) \to s$

Problem

Example 1.3 Determine which of the following compound propositions are tautologies and which of them are contradictions, using truth tables:

(a) $(\urcorner q \wedge (p \to q)) \to \urcorner p$

(b) $((p \to q) \wedge (q \to r)) \to (p \to r)$

(c) $\urcorner(q \to r) \wedge r \wedge (p \to q)$

(d) $((p \vee q) \wedge (p \to r) \wedge (q \to r)) \to r.$

Problem

Prove the following equivalences:

(a) $\neg(p \to q) \equiv p \wedge \neg q$

(b) $(p \wedge q) \vee (p \wedge \neg q) \equiv p$

(c) $(p \vee q) \wedge (p \vee \neg q) \equiv p$

(d) $\neg(p \leftrightarrow q) \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$

2. Check whether the propositions are logically equivalent or not,

a) $\neg p \to (q \to r)$ and $q \to (p \vee r)$

b) $p \to (q \wedge r)$ and $(p \to q) \wedge (p \to r)$ c) $[p \to (q \vee r)] \Leftrightarrow [\neg r \to (p \to q)]$

Duality

1.2.1 Duality

Let S be a statement. If S contain no logical connectives other than ∨ and ∧, then the dual of S, denoted by S^d, is the statement obtained from S by replacing ∨ and ∧ by ∧ and ∨ respectively and T_0 by F_0 and F_0 by T_0.

For example, let $S = (p \wedge \neg p) \vee (r \wedge T_0)$

The dual of the statement is $S^d = (p \vee \neg p) \wedge (r \vee F_0)$.

Principle of Duality

Let S and T be statements that contain no logical connectives other than ∨ and ∧. If $S \Leftrightarrow T$ then $S^d \Leftrightarrow T^d$.

Laws of Logic

Equivalences	Name
$p \wedge T \Leftrightarrow p$ $p \vee F \Leftrightarrow p$	Identity law
$p \vee T \Leftrightarrow T$ $p \wedge F \Leftrightarrow F$	Dominent law
$p \vee p \Leftrightarrow \overline{p}$ $p \wedge p \Leftrightarrow p$	Idempotent law
$p \vee q \Leftrightarrow q \vee p$ $p \wedge q \Leftrightarrow q \wedge p$	Commutative law
$(p \vee q) \vee r \Leftrightarrow p \vee (q \vee r)$ $(p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r)$	Associative law

Laws of Logic(Continued...)

Equivalences	Name
$(p \vee q) \wedge r \Leftrightarrow (p \wedge r) \vee (q \wedge r)$ $(p \wedge q) \vee r \Leftrightarrow (p \vee r) \wedge (q \vee r)$	Distributive law
$(p \vee q) \wedge p \Leftrightarrow p$ $(p \wedge q) \vee p \Leftrightarrow p$	Absorbtion law
$\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$ $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$	De morgan's law
$\neg p \wedge p \Leftrightarrow F$ $\neg p \vee p \Leftrightarrow T$ $\neg(\neg p) \Leftrightarrow p$	Negation law

Laws of Logic(Continued...)

Using equivalences, we can *define* operators in terms of other operators.

- Exclusive or: $p \oplus q \Leftrightarrow (p \vee q) \wedge \neg(p \wedge q)$
  $p \oplus q \Leftrightarrow (p \wedge \neg q) \vee (q \wedge \neg p)$
- Implies: $p \rightarrow q \Leftrightarrow \neg p \vee q$
- Biconditional: $p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$
  $p \leftrightarrow q \Leftrightarrow \neg(p \oplus q)$

Laws of Logic(Continued...)

Implications
$p \rightarrow q \Leftrightarrow \neg p \vee q$
$p \rightarrow q \Leftrightarrow \neg q \rightarrow \neg p$
$\neg(p \rightarrow q) \Leftrightarrow p \wedge \neg q$
$p \vee q \Leftrightarrow \neg p \rightarrow q$
$p \wedge q \Leftrightarrow \neg(p \rightarrow \neg q)$
$(p \rightarrow q) \wedge (p \rightarrow r) \Leftrightarrow p \rightarrow (q \wedge r)$
$(p \rightarrow q) \vee (p \rightarrow r) \Leftrightarrow p \rightarrow (q \vee r)$
$(p \rightarrow r) \wedge (q \rightarrow r) \Leftrightarrow (p \vee q) \rightarrow r$
$(p \rightarrow r) \vee (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$

Laws of Logic(Continued...)

Biconditions
$p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$
$p \leftrightarrow q \Leftrightarrow \neg p \leftrightarrow \neg q$
$p \leftrightarrow q \Leftrightarrow (p \wedge q) \vee (\neg p \wedge \neg q)$
$\neg(p \leftrightarrow q) \Leftrightarrow p \leftrightarrow \neg q$

Substitution Rules

- First Substitution Rule

Suppose that the compound statement S is a tautology. If p is a primitive statement that appears in S and we replace each occurrence of p by a statement q, then the resulting compound statement D is also a tautology.

Eg: S: $\neg(p \vee q) \leftrightarrow (\neg p \wedge \neg q)$ is a tautology . If we replace p by $r \wedge s$, it follows from the first substitution rule that

S: $\neg((r \wedge s) \vee q) \leftrightarrow (\neg(r \wedge s) \wedge \neg q)$ is also a tautology

Substitution Rules (Continued...)

- Second Substitution Rule

Let S be a compound statement where p is a primitive statement that appears in S and let q be a statement such that $p \Leftrightarrow q$. Suppose that in S if we replace one or more occurrences of p by q, the resulting statement is D and $S \Leftrightarrow D$.

Eg: $S:(p \longrightarrow q) \longrightarrow r$.

Since $(p \longrightarrow q) \Leftrightarrow (\neg p \vee q)$.

$D:((\neg p \vee q) \longrightarrow r$.

Then by second substitution rule $S \Leftrightarrow D$

Problems

Write the Dual of a) $q \rightarrow r$ b) $p \rightarrow q \vee r$ c) $p \leftrightarrow q$ where p,q and r are primitive statements.

Write the Dual of the logical equivalence $(\neg p \vee q) \wedge (p \wedge (p \wedge q) \Leftrightarrow (p \wedge q)$ where p,q and r are primitive statements.

Ans:

Solution:

a) $q \to r \Leftrightarrow \neg q \vee r$. The duel of the statement is $(q \to r)^d = \neg q \wedge r$

b) $p \to (q \vee r) \Leftrightarrow \neg p \vee (q \vee r)$. The dual is $\neg p \wedge (q \wedge r)$

c) $p \leftrightarrow q \Leftrightarrow (p \to q) \wedge (q \to p) \Leftrightarrow (\neg p \vee q) \wedge (\neg q \vee p)$

So, the duel of the statement is $(p \leftrightarrow q)^d = (\neg p \wedge q) \vee (\neg q \wedge p)$.

$$(\neg p \wedge q) \vee (p \vee (p \vee q) \Leftrightarrow (p \vee q)$$

With out using Truth table:

(i) Show that $(p \vee q) \wedge \neg p \Leftrightarrow (\neg p \wedge q)$

(ii) Show that $(p \vee q) \wedge \neg(\neg p \wedge q) \Leftrightarrow p$

(iii) Show that $\neg(p \vee (\neg p \wedge q)) \Leftrightarrow \neg p \wedge \neg q$

(iv) Show that $\neg(\neg((p \vee q) \wedge r) \vee \neg q) \Leftrightarrow q \wedge r$

(i)

$S \Leftrightarrow (p \vee q) \wedge \neg p$	Reasons
$\Leftrightarrow (p \vee q) \wedge \neg p$	Given
$\Leftrightarrow (p \wedge \neg p) \vee (q \wedge \neg p)$	Distributive law
$\Leftrightarrow F \vee (\neg p \wedge q)$	Negation law,Commutative law
$\Leftrightarrow \neg p \wedge q$	Identity law

(ii)

$S \Leftrightarrow (p \vee q) \wedge \neg(\neg p \wedge q)$	Reasons
$\Leftrightarrow (p \vee q) \wedge \neg(\neg p \wedge q)$	Given
$\Leftrightarrow (p \vee q) \wedge (\neg\neg p \vee \neg q)$	De Morgan's law
$\Leftrightarrow (p \vee q) \wedge (p \vee \neg q)$	Negation law
$\Leftrightarrow p \vee (q \wedge \neg q)$	Distributive law
$\Leftrightarrow p \vee F$	Negation law
$\Leftrightarrow p$	Identity law

(iii)

$S \Leftrightarrow \neg(p \vee (\neg p \wedge q))$	Reasons
$\Leftrightarrow \neg(p \vee (\neg p \wedge q))$	Given
$\Leftrightarrow \neg p \wedge \neg(\neg p \wedge q))$	De Morgan's law
$\Leftrightarrow \neg p \wedge (p \vee \neg q))$	De Morgan's law
$\Leftrightarrow (\neg p \wedge p) \vee (\neg p \wedge \neg q)$	Distributive law
$\Leftrightarrow F \vee (\neg p \wedge \neg q)$	Negation law
$\Leftrightarrow \neg p \wedge \neg q$	Identity law

(iv)

$S \Leftrightarrow \neg(\neg((p \vee q) \wedge r) \vee \neg q) \Leftrightarrow q \wedge r$	Reasons
$\Leftrightarrow \neg(\neg((p \vee q) \wedge r) \vee \neg q)$	Given
$\Leftrightarrow ((p \vee q) \wedge r) \wedge q$	De Morgan's law
$\Leftrightarrow (p \vee q) \wedge (r \wedge q)$	Associative law
$\Leftrightarrow (p \wedge (r \wedge q)) \vee (q \wedge (r \wedge q))$	Distributive law
$\Leftrightarrow (p \wedge (r \wedge q)) \vee (r \wedge q)$	Idempotent law
$\Leftrightarrow r \wedge q$	Absorption law

Prove that $[(p \vee q) \wedge ((p \vee \neg q)] \vee q \Leftrightarrow p \vee q$

Solution:

$[(p \vee q) \wedge ((p \vee \neg q)] \vee q$	
$\Leftrightarrow [(p \vee (q \wedge \neg q)] \vee q$	Distributive Law
$\Leftrightarrow (p \vee F_0) \vee q$	Inverse Law
$\Leftrightarrow (p \vee q)$	Identity Law

Prove that the statements are logically equivalent without using truth table

$$(p \vee q) \wedge \neg(\neg p \wedge q) \Leftrightarrow p$$

Solution:

$(p \vee q) \wedge \neg(\neg p \wedge q)$	
$\Leftrightarrow (p \vee q) \wedge (\neg\neg p \vee \neg q)$	Demorgan's Law
$\Leftrightarrow (p \vee q) \wedge (p \vee \neg q)$	Law of double negation
$\Leftrightarrow p \vee (q \wedge \neg q)$	Distributive Law
$\Leftrightarrow p \vee F_0$	Inverse Law
$\Leftrightarrow p$	Identity Law

Prove that $(p \to q) \wedge (r \to q) \Leftrightarrow (p \vee r) \to q$

Solution:

$(p \to q) \wedge (r \to q)$	
$\Leftrightarrow (\neg p \vee q) \wedge (\neg r \vee q)$	Since $(p \to q) \Leftrightarrow (\neg p \vee q)$
$\Leftrightarrow (\neg p \wedge \neg r) \vee q$	Distributive Law
$\Leftrightarrow \neg(p \vee r) \vee q$	Demorgan's Law
$\Leftrightarrow p \vee r \to q$	Since $(p \to q) \Leftrightarrow (\neg p \vee q)$

Prove that $(p \vee q) \wedge (\neg(\neg p \wedge q)) \Leftrightarrow p$

Solution:

$(p \vee q) \wedge (\neg(\neg p \wedge q)) \Leftrightarrow p$	
$\Leftrightarrow (p \vee q) \wedge (\neg\neg p \vee \neg q))$	Demorgan's Law
$\Leftrightarrow (p \vee q) \wedge (p \vee \neg q))$	Law of double negation
$\Leftrightarrow (p \vee (q \wedge \neg q)$	Distributive Law
$\Leftrightarrow (p \vee F_0)$	Inverse Law
$\Leftrightarrow p$	Identity Law

Prove that $(p \rightarrow q) \wedge \neg q \Leftrightarrow \neg(p \vee q)$

Solution:

$(p \rightarrow q) \wedge \neg q$	
$\Leftrightarrow (\neg p \vee q) \wedge \neg q$	Since $(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$
$\Leftrightarrow (\neg p \wedge \neg q) \vee (q \wedge \neg q)$	Distributive Law
$\Leftrightarrow (\neg p \wedge \neg q) \vee F_0$	Inverse Law
$\Leftrightarrow (\neg p \wedge \neg q)$	Identity Law
$\Leftrightarrow \neg(p \vee q)$	Demorgan's Law

Prove that $(p \rightarrow q) \wedge \neg q \Leftrightarrow \neg(p \vee q)$

Solution:

$(p \rightarrow q) \wedge \neg q$	
$\Leftrightarrow (\neg p \vee q) \wedge \neg q$	Since $(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$
$\Leftrightarrow (\neg p \wedge \neg q) \vee (q \wedge \neg q)$	Distributive Law
$\Leftrightarrow (\neg p \wedge \neg q) \vee F_0$	Inverse Law
$\Leftrightarrow (\neg p \wedge \neg q)$	Identity Law
$\Leftrightarrow \neg(p \vee q)$	Demorgan's Law

Without using truth table, show that $p \to (q \to r) \Leftrightarrow p \to (\neg q \vee r) \Leftrightarrow (p \wedge q) \to r$

Solution:

$p \to (q \to r)$	
$\Leftrightarrow p \to (\neg q \vee r)$	Since $(p \to q) \Leftrightarrow (\neg p \vee q)$
$\Leftrightarrow \neg p \vee (\neg q \vee r)$	Since $(p \to q) \Leftrightarrow (\neg p \vee q)$
$\Leftrightarrow (\neg p \vee \neg q) \vee r$	Associative Law
$\Leftrightarrow \neg(p \wedge q) \vee r$	Demorgan's Law
$\Leftrightarrow (p \wedge q) \to r$	Since $(p \to q) \Leftrightarrow (\neg p \vee q)$

Prove that $\neg[\neg((p \vee q) \wedge r) \vee \neg q] \Leftrightarrow q \wedge r$

Solution:

$\neg[\neg((p \vee q) \wedge r) \vee \neg q]$	
$\Leftrightarrow [\neg\neg((p \vee q) \wedge r) \wedge \neg\neg q]$	Demorgan's Law
$\Leftrightarrow [((p \vee q) \wedge r)] \wedge q)$	Law of double negation
$\Leftrightarrow (p \vee q) \wedge (r \wedge q)$	Associative Law
$\Leftrightarrow (p \vee q) \wedge (q \wedge r)$	Commutative Law
$\Leftrightarrow ((p \vee q) \wedge q) \wedge r)$	Associative Law
$\Leftrightarrow (q \wedge (p \vee q)) \wedge r)$	Commutative Law
$\Leftrightarrow q \wedge r$	Absorption Law

Prove that $(p \vee q) \wedge (\neg p \wedge (\neg p \wedge q)) \Leftrightarrow (\neg p \wedge q)$

Solution:

$(p \vee q) \wedge (\neg p \wedge (\neg p \wedge q))$	
$\Leftrightarrow (p \vee q) \wedge ((\neg p \wedge \neg p) \wedge q))$	Associative Law
$\Leftrightarrow (p \vee q) \wedge (\neg p \wedge q)$	Idempotent Law
$\Leftrightarrow [p \wedge (\neg p \wedge q)] \vee [q \wedge (\neg p \wedge q)]$	Distributive Law
$\Leftrightarrow [(p \wedge \neg p) \wedge q] \vee [(q \wedge \neg p) \wedge q)]$	Associative Law
$\Leftrightarrow ((p \wedge \neg p) \vee (q \wedge \neg p)) \wedge q)$	Distributive Law
$\Leftrightarrow (F_0 \vee (q \wedge \neg p)) \wedge q)$	Inverse Law
$\Leftrightarrow (q \wedge \neg p) \wedge q)$	Identity Law
$\Leftrightarrow (\neg p \wedge q) \wedge q)$	Commutative Law
$\Leftrightarrow \neg p \wedge (q \wedge q)$	Associative Law
$\Leftrightarrow (\neg p \wedge q)$	Idempotent Law

Prove that $(\neg p \wedge (\neg q \wedge r) \vee (q \wedge r) \vee (p \wedge r) \Leftrightarrow r$

Solution:

$(\neg p \wedge (\neg q \wedge r) \vee (q \wedge r) \vee (p \wedge r)$	
$\Leftrightarrow [(\neg p \wedge \neg q) \wedge r)] \vee (q \wedge r) \vee (p \wedge r)$	Associative Law
$\Leftrightarrow [\neg(p \vee q) \wedge r)] \vee (q \wedge r) \vee (p \wedge r)$	Demorgan's Law
$\Leftrightarrow [\neg(p \vee q) \wedge r)] \vee [(q \vee p) \wedge r]$	Distributive Law
$\Leftrightarrow [\neg(p \vee q) \vee (q \vee p)] \wedge r$	Distributive Law
$\Leftrightarrow [\neg(p \vee q) \vee (p \vee q)] \wedge r$	Commutative Law
$\Leftrightarrow T_0 \wedge r$	Inverse Law
$\Leftrightarrow r$	Identity Law

Logical Implication: ($\Rightarrow$)

- A statement p is said to logically imply another statement q if and only if $p \rightarrow q$ is a tautology. This is denoted by $p \Rightarrow q$ and read as p logically implies q.

ie. $p \Rightarrow q$ iff $p \rightarrow q$ is a tautology

Argument

- An argument in Propositional logic is a sequence of statements (propositions).
- All statements except the final statement in the argument are called premises and the final statement is called the conclusion.

Representing an argument

- An argument is sometimes written as follows:

$$\begin{array}{l} p_1 \\ p_2 \\ \vdots \\ p_n \\ \hline \therefore q \end{array}$$

premises

conclusion

Example

- Premises

  "If you have a current password, you can log onto the computer network."

  "You have a current password."

- Conclusion

  "Therefore, you can log onto the computer network."

Validity of an argument

- An argument is valid if and only if the truth of all its premises implies that the conclusion is true.
- If we have an argument with premises p_1, p_2, p_3, …………,p_n and conclusion q.

This argument is valid when

$(p_1 \wedge p_2 \wedge p_3 \wedge \ldots\ldots\ldots\ldots \wedge p_n) \rightarrow q$ is a tautology.

Fallacy

- An argument which is not valid is called a fallacy.
- How do we show that an argument is valid?
  - We can use a truth table. By showing that $(p_1 \wedge p_2 \wedge p_3 \wedge \ldots\ldots\ldots\ldots \wedge p_n) \rightarrow q)$ is a tautology or whenever the premises are true, the conclusion must also be true.
  - We can show that $(p_1 \wedge p_2 \wedge p_3 \wedge \ldots\ldots\ldots\ldots \wedge p_n) \rightarrow q)$ is a tautology using some rules of inference.

Why use rules of inference?

- Constructing a truth table is time consuming!
- If we have n propositions, what is the size of the truth table? 2^n, which means that the table doubles in size with every proposition. This is a tedious(long and boring) approach.

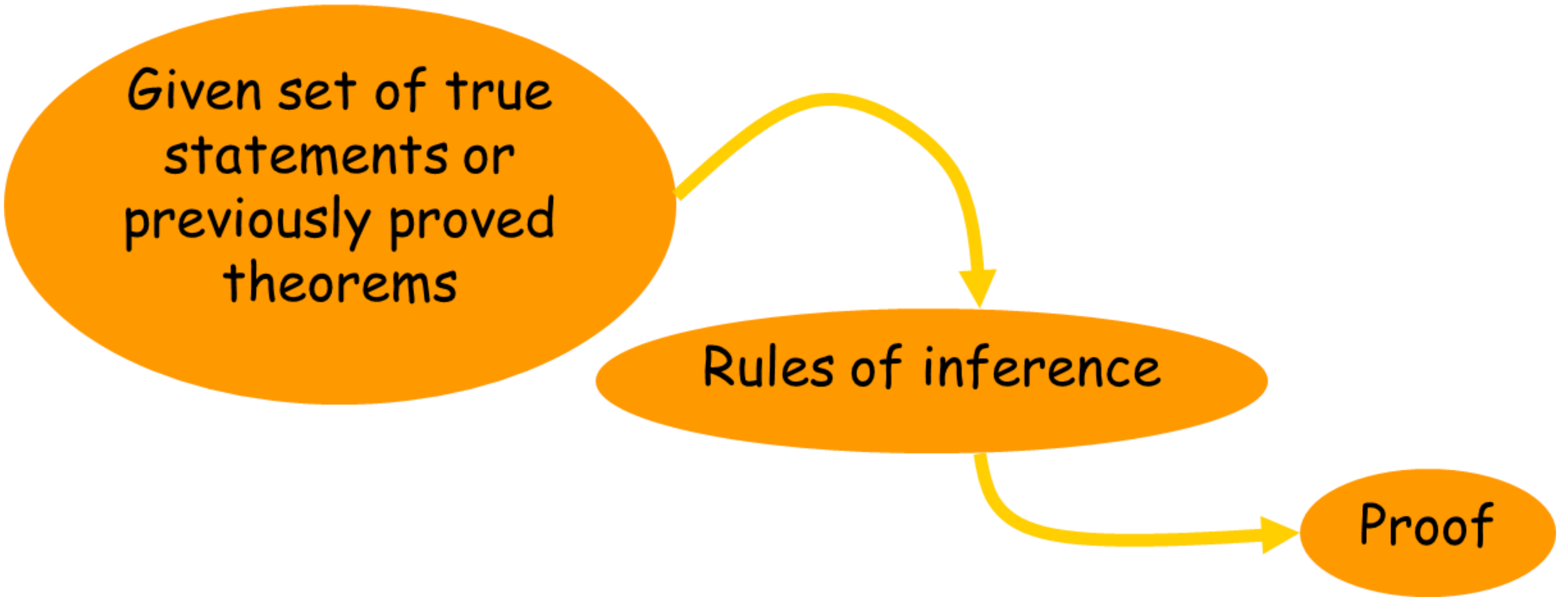
Rule of Inference	Related Logical Implication	Name of Rule
1) $\begin{array}{l} p \\ p \to q \\ \hline \therefore q \end{array}$	$[p \wedge (p \to q)] \to q$	Rule of Detachment (Modus Ponens)
2) $\begin{array}{l} p \to q \\ q \to r \\ \hline \therefore p \to r \end{array}$	$[(p \to q) \wedge (q \to r)] \to (p \to r)$	Law of the Syllogism
3) $\begin{array}{l} p \to q \\ \neg q \\ \hline \therefore \neg p \end{array}$	$[(p \to q) \wedge \neg q] \to \neg p$	Modus Tollens
4) $\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$		Rule of Conjunction

5)	$\begin{array}{l} p \vee q \\ \underline{\neg p} \\ \therefore q \end{array}$	$[(p \vee q) \wedge \neg p] \rightarrow q$	Rule of Disjunctive Syllogism
6)	$\begin{array}{l} \underline{\neg p \rightarrow F_0} \\ \therefore p \end{array}$	$(\neg p \rightarrow F_0) \rightarrow p$	Rule of Contradiction
7)	$\begin{array}{l} \underline{p \wedge q} \\ \therefore p \end{array}$	$(p \wedge q) \rightarrow p$	Rule of Conjunctive Simplification
8)	$\begin{array}{l} \underline{p} \\ \therefore p \vee q \end{array}$	$p \rightarrow p \vee q$	Rule of Disjunctive Amplification

9)	$p \wedge q$ $p \rightarrow (q \rightarrow r)$ $\therefore r$	$[(p \wedge q) \wedge [p \rightarrow (q \rightarrow r)]] \rightarrow r$	Rule of Conditional Proof
10)	$p \rightarrow r$ $q \rightarrow r$ $\therefore (p \vee q) \rightarrow r$	$[(p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow [(p \vee q) \rightarrow r]$	Rule for Proof by Cases
11)	$p \rightarrow q$ $r \rightarrow s$ $p \vee r$ $\therefore q \vee s$	$[(p \rightarrow q) \wedge (r \rightarrow s) \wedge (p \vee r)] \rightarrow (q \vee s)$	Rule of the Constructive Dilemma
12)	$p \rightarrow q$ $r \rightarrow s$ $\neg q \vee \neg s$ $\therefore \neg p \vee \neg r$	$[(p \rightarrow q) \wedge (r \rightarrow s) \wedge (\neg q \vee \neg s)] \rightarrow (\neg p \vee \neg r)$	Rule of the Destructive Dilemma

Proofs – How do we know?

- A theorem is a statement that can be shown to be true.
- A proof is the means of doing so.



Proofs – How do we know?

- The following statements are true:
  - If I have taken MAT 203, then I am allowed to take MAT 206
  - I have taken MAT 203
- What do we know to be true?
  - I am allowed to take MAT 206

What rule of inference can we use to justify it?

What rules we study

- Modus Ponens
- Modus Tollens
- Disjunctive Amplification
- Conjunctive Simplification
- Hypothetical Syllogism
- Proof by Cases
- Conjunction
- Disjunctive Syllogism
- Contradiction

Modus Ponens

I have taken MAT 203.
If I have taken MAT 203, then I am allowed to take MAT 206.

$\therefore$ I am allowed to take MAT 206.

$$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

Tautology:

$$(p \wedge (p \rightarrow q)) \rightarrow q$$

Inference Rule:

Modus Ponens

Modus Tollens

I am not allowed to take MAT 206.
If I have taken MAT 203 , then I am allowed to take MAT 206.

$\therefore$ I have not taken MAT 203

$$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$$

Tautology:

$$(\neg q \land (p \rightarrow q)) \rightarrow \neg p$$

Inference Rule:

Modus Tollens

Disjunctive Amplification (Addition)

I am not a great skater.

$\therefore$ I am not a great skater or I am tall.

$$\frac{p}{\therefore\ p \vee q}$$

Tautology:

$p \rightarrow (p \vee q)$

$$\frac{q}{\therefore\ p \vee q}$$

Tautology:

$q \rightarrow (p \vee q)$

Inference Rule:

Disjunctive Amplification

Conjunctive Simplification

I am not a great skater, and you are sleepy.

$\therefore$ you are sleepy.

$$\frac{p \wedge q}{\therefore\ p}$$

Tautology:

$(p \wedge q) \to p$

Inference Rule:

Conjunctive Simplification

$$\frac{p \wedge q}{\therefore\ q}$$

Tautology:

$(p \wedge q) \to q$

Hypothetical Syllogism

If you are an athlete, you are always hungry.

If you are always hungry, you have a snickers in your backpack.

$\therefore$ If you are an athlete, you have a snickers in your backpack.

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore\ p \rightarrow r \end{array}$$

Tautology:

$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

Inference Rule:

Hypothetical Syllogism

Proof by Cases

If you are an athlete, you have a snickers in your backpack.

If you are always hungry, you have a snickers in your backpack.

$\therefore$ If you are an athlete and you are always hungry, you have a snickers in your backpack.

$$p \rightarrow r$$
$$q \rightarrow r$$
$$\overline{\therefore (p \vee q) \rightarrow r}$$

Tautology:

$((p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow (p \vee q) \rightarrow r$

Inference Rule:

Proof by cases

Conjunction

You are an athlete

You have a snickers in your backpack.

You are always hungry, and you have a snickers in your backpack.

$$\begin{array}{c} p \\ q \\ \hline \therefore\ p \wedge q \end{array}$$

Inference Rule:

Conjunction

Disjunctive Syllogism

I am a great eater, or I am a great skater.

I am not a great skater.

$\therefore$ I am a great eater!

$p \vee q$

$\neg q$

$\therefore p$

Tautology:

$((p \vee q) \wedge \neg q) \rightarrow p$

Inference Rule:

Disjunctive Syllogism

Contradiction

$\neg p \rightarrow F$

$\therefore p$

Tautology:

$(\neg p \rightarrow F) \rightarrow p$

Inference Rule:

Contradiction

Another view on what we are doing

You might think of this as some sort of game.

You are given some statement, and you want to see if it is a valid argument and true

You translate the statement into argument form using propositional variables, and make sure you have the premises right, and clear what is the conclusion

You then want to get from premises/hypotheses (A) to the conclusion (B) using the rules of inference.

So, get from A to B using as "moves" the rules of inference

Rules of inference:

Rule P: A premises can be introduced at any step of derivation.

Rule T: A formula can be introduced provided it is Tautologically implied by previously introduced formulas in the derivation.

Rule CP: If the conclusion is of the form $r \to s$ then we include r as an additional premises and derive s.

Indirect method: We use negation of the conclusion as an additional premise and try to arrive a contradiction.

Inconsistent: A set of premises are inconsistent provided their conjunction implies a contradiction.

We also introduce another rule of inference below.

(CP): If the conclusion is a conditional statement $(A \to B)$ then include A as an additional premise, and then derive B.

We also include the following terminology:

- Indirect method: Suppose we want to conclude A from a given set of premises. We then include $(\neg A)$ as an additional premise, and then derive a wff B where B is a contradiction.
- Inconsistent set of premises: A set of premises is said to be *inconsistent* if we can conclude a wff B, where B is a contradiction.

Verify the validity of the arguments using Truth table.

1) $p \wedge [(p \wedge r) \to s] \to q$

2) $(\neg p) \wedge (p \vee q) \to p \wedge q$

3) $(\neg p) \wedge (p \vee q) \to q$

4) $p \wedge [(p \to q) \wedge r] \to (p \vee q) \to r$

4) $\{[(p \wedge q) \to r] \wedge \neg q \wedge (p \to \neg r)\} \to (\neg p \vee \neg q)$

5) $\{[p \vee (q \vee r)] \wedge \neg q\} \to p \vee r$

Problems

1. It is not sunny this afternoon and it is colder than yesterday.
2. If we go swimming it is sunny.
3. If we do not go swimming, then we will take a canoe trip.
4. If we take a canoe trip, then we will be home by sunset.
5. We will be home by sunset

p	It is sunny this afternoon
q	It is colder than yesterday
r	We go swimming
s	We will take a canoe trip
t	We will be home by sunset (the conclusion)

propositions

1. $\neg p \wedge q$
2. $r \rightarrow p$
3. $\neg r \rightarrow s$
4. $s \rightarrow t$
5. t

hypotheses

Problems

	Step	Reason
1.	$\neg p \wedge q$	Hypothesis
2.	$\neg p$	Simplification using (1)
3.	$r \rightarrow p$	Hypothesis
4.	$\neg r$	Modus tollens using (2) and (3)
5.	$\neg r \rightarrow s$	Hypothesis
6.	s	Modus ponens using (4) and (5)
7.	$s \rightarrow t$	Hypothesis
8.	t	Modus ponens using (6) and (7)

1. *Anna is skiing or it is not snowing.*
2. *It is snowing or Bart is playing hockey.*
3. *Consequently, Anna is skiing, or Bart is playing hockey.*

hypotheses

1. $p \vee \neg r$
2. $r \vee q$

propositions

p	Anna is skiing
q	Bart is playing hockey
r	it is snowing

$$\begin{array}{l} p \vee q \\ \underline{\neg p \vee r} \\ \therefore q \vee r \end{array}$$

← Resolution rule

Consequently, Anna is skiing, or Bart is playing hockey

Problems

Here's what you know:

Ellen is a math major or a CS major.
If Ellen does not like discrete math, she is not a CS major.
If Ellen likes discrete math, she is smart.
Ellen is not a math major.

Can you conclude Ellen is smart?

$$M \vee C$$
$$\neg D \rightarrow \neg C$$
$$D \rightarrow S$$
$$\neg M$$

1. $M \vee C$	Given
2. $\neg D \rightarrow \neg C$	Given
3. $D \rightarrow S$	Given
4. $\neg M$	Given
5. C	DS (1,4)
6. D	MT (2,5)
7. S	MP (3,6)

Ellen is smart!

Our first example demonstrates the validity of the argument

$$\begin{array}{l} p \to r \\ \neg p \to q \\ q \to s \\ \hline \therefore \neg r \to s \end{array}$$

Steps	Reasons
1) $p \to r$	Premise
2) $\neg r \to \neg p$	Step (1) and $p \to r \iff \neg r \to \neg p$
3) $\neg p \to q$	Premise
4) $\neg r \to q$	Steps (2) and (3) and the Law of the Syllogism
5) $q \to s$	Premise
6) $\therefore \neg r \to s$	Steps (4) and (5) and the Law of the Syllogism

A second way to validate the given argument proceeds as follows.

Steps	Reasons
1) $p \to r$	Premise
2) $q \to s$	Premise
3) $\neg p \to q$	Premise
4) $p \lor q$	Step (3) and $(\neg p \to q) \iff (\neg\neg p \lor q) \iff (p \lor q)$, where the second logical equivalence follows by the Law of Double Negation
5) $r \lor s$	Steps (1), (2), and (4) and the Rule of the Constructive Dilemma
6) $\therefore \neg r \to s$	Step (5) and $(r \lor s) \iff (\neg\neg r \lor s) \iff (\neg r \to s)$, where the Law of Double Negation is used in the first logical equivalence

Establish the validity of the argument

$$\begin{array}{l} p \to q \\ q \to (r \wedge s) \\ \neg r \vee (\neg t \vee u) \\ p \wedge t \\ \hline \therefore u \end{array}$$

Steps	Reasons
1) $p \to q$	Premise
2) $q \to (r \wedge s)$	Premise
3) $p \to (r \wedge s)$	Steps (1) and (2) and the Law of the Syllogism
4) $p \wedge t$	Premise
5) p	Step (4) and the Rule of Conjunctive Simplification
6) $r \wedge s$	Steps (5) and (3) and the Rule of Detachment
7) r	Step (6) and the Rule of Conjunctive Simplification
8) $\neg r \vee (\neg t \vee u)$	Premise
9) $\neg(r \wedge t) \vee u$	Step (8), the Associative Law of $\vee$, and DeMorgan's Laws
10) t	Step (4) and the Rule of Conjunctive Simplification
11) $r \wedge t$	Steps (7) and (10) and the Rule of Conjunction
12) $\therefore u$	Steps (9) and (11), the Law of Double Negation, and the Rule of Disjunctive Syllogism

If the band could not play rock music or the refreshments were not delivered on time, then the New Year's party would have been canceled and Alicia would have been angry. If the party were canceled, then refunds would have had to be made. No refunds were made.

Therefore the band could play rock music.

First we convert the given argument into symbolic form by using the following statement assignments:

p: The band could play rock music.
q: The refreshments were delivered on time.
r: The New Year's party was canceled.
s: Alicia was angry.
t: Refunds had to be made.

The argument above now becomes

$$\begin{array}{l} (\neg p \vee \neg q) \rightarrow (r \wedge s) \\ r \rightarrow t \\ \neg t \\ \hline \therefore p \end{array}$$

We can establish the validity of this argument as follows.

Steps	Reasons
1) $r \rightarrow t$	Premise
2) $\neg t$	Premise
3) $\neg r$	Steps (1) and (2) and Modus Tollens
4) $\neg r \vee \neg s$	Step (3) and the Rule of Disjunctive Amplification
5) $\neg(r \wedge s)$	Step (4) and DeMorgan's Laws
6) $(\neg p \vee \neg q) \rightarrow (r \wedge s)$	Premise
7) $\neg(\neg p \vee \neg q)$	Steps (6) and (5) and Modus Tollens
8) $p \wedge q$	Step (7), DeMorgan's Laws, and the Law of Double Negation
9) $\therefore p$	Step (8) and the Rule of Conjunctive Simplification

Rita is baking a cake. If Rita is baking a cake, then she is not practicing her flute. If Rita is not practicing her flute, then her father will not buy her a car. Therefore, Rita's father will not buy her a car. Check the validity of the statement.

Solution:

The given propositions are

p: Rita is baking a cake

q: Rita is practicing her flute

r: Rita's father will buy her a car

The argument is

$$p$$

$$p \to \neg q$$

$$\neg q \to \neg r$$

....

$$\therefore \neg r$$

	Steps	Reason
1	p	Premises
2	$p \rightarrow \neg q$	Premises
3	$\neg q \rightarrow \neg r$	Premises
4	$p \rightarrow \neg r$	Step (2), (3) Law of Syllogism
5	$\neg r$	Step (1), (4) Modus Ponens

The conclusion follows from the premises. $\therefore$ the argument is valid.

Establish the validity of the argument by the method of proof by contradiction

$$\neg p \leftrightarrow q$$
$$q \rightarrow r$$
$$\neg r$$
$$\text{.......................}$$
$$\therefore\ p$$

Solution:

	Steps	**Reason**
1	$\neg p \leftrightarrow q$	Premises
2	$(\neg p \rightarrow q) \wedge (q \rightarrow \neg p)$	Step(1)and $(p \leftrightarrow q) \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$
3	$\neg p \rightarrow q$	Step (2) Rule of conjunctive Simplification
4	$q \rightarrow r$	Premises
5	$\neg p \rightarrow r$	Step (3),(4) Law of the Syllogism
6	$\neg p$	Premises (assume negation of conclusion)
7	r	Step (5), (6) and Modus Ponens
8	$\neg r$	Premises
9	$r \wedge \neg r$	Step (7), (8) Rule of Conjunction
10	F_0	Step (9) Inverse law
11	$\therefore\ p$	Step (6),(10) Method of proof by Contradiction

Establish the validity of the argument

$$p \to (q \to r)$$
$$p \vee \neg s$$
$$q$$
$$\overline{\qquad\qquad}$$
$$\therefore s \to r$$

Solution:

	Steps	**Reason**
1	$p \to (q \to r)$	Premises
2	$p \vee \neg s$	Premises
3	q	Premises
4	s	Premises (assume)
5	$\neg s \vee p$	Step (2) commutative property
6	$s \to p$	Step (5) $p \to q \Leftrightarrow \neg p \vee q$
7	$s \to (q \to r)$	Step (1), (6) Law of the Syllogism
8	$\neg s \vee (q \to r)$	Step (7) $p \to q \Leftrightarrow \neg p \vee q$
9	$\neg s \vee (\neg q \vee r)$	Step (8) $p \to q \Leftrightarrow \neg p \vee q$
10	$(\neg s \vee \neg q) \vee r$	Step (9) Associative Property
11	$\neg(s \wedge q) \vee r$	Step (10) Demorgan's Law
12	$(s \wedge q) \to r$	Step (11) $p \to q \Leftrightarrow \neg p \vee q$
13	$s \wedge q$	Step (3) (4) Rule of Conjunction
14	r	Step (12), (13) Modus Ponens
15	$\therefore s \to r$	Step (4), (14)

PREDICATE LOGIC

Predicate:

- A predicate is a statement that contains variables (predicate variables), and they may be true or false depending on the values of these variables.

- *ie,* A predicate is a sentence that contains a finite number of variables and becomes a statement when specific values are substituted for the variables

Example:

- $P(x)$ = "x^2 is greater than x " is a predicate. If we choose $x = 1$, $P(1)$ is " 1 is greater than 1 ", which is a proposition (always false).

Open statement

- *A declarative sentence is an open statement if*

1. it contains one or more variables, and

2. it is not a statement, but

3. it becomes a statement when the variables in it are replaced by certain allowable choices.

- *x, y) : "x = y"*

- *q(x, y) : "x is brother of y"*

Universe of Discourse

- *If p(x) is an open statement containing the variable x, then the domain of the variable x (i.e. scope of the variable x) is called as the Universe of discourse or universe.*

- *It can be the set of real numbers, the set of integers, the set of all cars on a parking lot, the set of all students in a classroom etc.*

Free and Bound Variables

- *The variables in open statements are called free variables.*

- *The variables which are quantified are called bound variables.*

Quantifiers

- *A quantifier is "an operator that limits the variables of a proposition"*

Two types:

- *Universal*
- *Existential*

Universal quantifiers

- *Represented by an upside-down A: ∀*
- *Universal quantifier states that the statements within its scope are true for every value of the specific variable.*
- *It is denoted by the symbol ∀ x p(x) is read as for every value of x, P(x) is true.*

Universal quantifiers

- *The universal quantifier is denoted by $\forall x$ and is read as "For all x", "For any x", "For each x" or "For every x".*
- *"For all x, y" , "For any x, y", "For every x, y" or "For all x and y" is denoted by $\forall x \forall y$ which can be abbreviated to $\forall x, y$.*

Example

- Let $P(x)$: "$x^2 + 2 = 7$" and the universe of discourse is the integers, then $\forall x\, P(x)$ is false.
- $Q(x)$ represents "$(x + 1)^2 = x^2 + 2x + 1$" then $\forall x\, Q(x)$ is true.

Existential quantifiers

- Represented by a backwards E: ∃
- The existential quantifier uses the words "For some x" , "for at least one x" or "there exists an x such that" .
- "For some x, p (x)" becomes $\exists\ x\ p(x)$, in symbolic form.
- "For some x, y , $q(x, y)$ " is $\exists x, y\ q\ (x, y)$.

Example

- *Let $P(x)$: "$x + 2 = 7$" with the integers as universe of discourse, then $\exists x\, P(x)$ is true, since there is indeed an integer, namely 5, such that $P(5)$ is a true statement.*

- *Let $Q(x)$: "$2x = 7$" with the integers as universe of discourse, then $\exists x\, Q(x)$ is false. But $\exists x\, Q(x)$ is true if we extend the universe of discourse to the rational numbers..*

Summary

Statement	When Is It True?	When Is It False?
$\exists x\ p(x)$	For some (at least one) a in the universe, $p(a)$ is true.	For every a in the universe, $p(a)$ is false.
$\forall x\ p(x)$	For every replacement a from the universe, $p(a)$ is true.	There is at least one replacement a from the universe for which $p(a)$ is false.
$\exists x\ \neg p(x)$	For at least one choice a in the universe, $p(a)$ is false, so its negation $\neg p(a)$ is true.	For every replacement a in the universe, $p(a)$ is true.
$\forall x\ \neg p(x)$	For every replacement a from the universe, $p(a)$ is false and its negation $\neg p(a)$ is true.	There is at least one replacement a from the universe for which $\neg p(a)$ is false and $p(a)$ is true.

Definition Contra positive, Converse & Inverse of $\forall x\,[p(x) \rightarrow q(x)]$

1) $\forall x\,[\neg q(x) \rightarrow \neg p(x)]$ Contrapositive

2) $\forall x\,[q(x) \rightarrow p(x)]$ Converse

3) $\forall x\,[\neg p(x) \rightarrow \neg q(x)]$ Inverse.

Logical Equivalences & Logical Implications

1) $\exists x\,[p(x) \wedge q(x)] \overset{\text{logically implies}}{\Longrightarrow} [\exists x\, p(x)] \wedge [\exists x\, q(x)]$

2) $\exists x\,[p(x) \vee q(x)] \overset{\text{logically equivalent}}{\Longleftrightarrow} [\exists x\, p(x) \vee \exists x\, q(x)]$

3) $\forall x\,[p(x) \wedge q(x)] \overset{\text{logically equivalent}}{\Longleftrightarrow} [\forall x\, p(x) \wedge \forall x\, q(x)]$

4) $[\forall x\, p(x) \vee \forall x\, q(x)] \overset{\text{logically implies}}{\Longrightarrow} \forall x\,[p(x) \vee q(x)]$

Negation of the Quantifiers

$$\neg\left[\forall x \, P(x)\right] \Longleftrightarrow \exists x \, \neg P(x)$$

$$\neg\left[\exists x \, P(x)\right] \Longleftrightarrow \forall x \, \neg P(x)$$

$$\neg\left[\forall x \, \neg P(x)\right] \Longleftrightarrow \exists x \, P(x)$$

$$\neg\left[\exists x \, \neg P(x)\right] \Longleftrightarrow \forall x \, P(x)$$

1. While representing statement using universal quantifieres ($\forall$) we commonly use the connective "if... then" ($\longrightarrow$)

2. White representing statement using existential quantifier ($\exists$) we commonly use the connective "and" "(but)" ($\wedge$)

Quantifier	Connective
$\forall$ For all, every	$\longrightarrow$ if .... then
$\exists$ There exists, atleast	$\wedge$ and, (but)

Negate and simplify

$(i)\ \forall x, [p(x) \wedge \neg q(x)]$

$(ii)\ \exists x, [(p(x) \vee q(x)) \rightarrow r(x)]$

$(iii)\ \forall x, [p(x) \vee q(x)]$

(i)

$$\begin{aligned}\neg\{\forall x,[p(x)\wedge\neg q(x)]\} &\Leftrightarrow \exists x,\neg[p(x)\wedge\neg q(x)]\\ &\Leftrightarrow \exists x,[\neg p(x)\vee\neg\neg q(x)]\\ &\Leftrightarrow \exists x,[\neg p(x)\vee q(x)]\end{aligned}$$

(ii)

$$\begin{aligned}\neg\{\exists x,[(p(x)\vee q(x))\to r(x)]\} &\Leftrightarrow \forall x,\neg[(p(x)\vee q(x))\to r(x)]\\ &\Leftrightarrow \forall x,\neg[\neg\,(p(x)\vee q(x))\vee r(x)]\\ &\Leftrightarrow \forall x,\neg[\neg\,(p(x)\vee q(x))]\wedge\neg r(x)\\ &\Leftrightarrow \forall x,[\neg\neg\,(p(x)\wedge\neg q(x))]\wedge\neg r(x)\end{aligned}$$

(iii)

$$\begin{aligned}\neg\,[\forall x,[p(x)\vee q(x)]] &\Leftrightarrow \exists x,\neg(px)\vee q(x)\\ &\Leftrightarrow \exists x,(\neg p(x)\wedge\neg q(x))\end{aligned}$$

Negate the statement: *If x is odd then x^2-1 is even*

$p(x)$: x is odd $\quad\quad q(x)$: $x^2 - 1$ is even.

$$\forall x\ [p(x) \rightarrow q(x)]$$

The negation of this statement is determined as follows:

$$\neg[\forall x\ (p(x) \rightarrow q(x))] \Leftrightarrow \exists x\ [\neg(p(x) \rightarrow q(x))]$$

$$\Leftrightarrow \exists x\ [\neg(\neg p(x) \vee q(x))] \Leftrightarrow \exists x\ [\neg\neg p(x) \wedge \neg q(x)]$$

$$\Leftrightarrow \exists x\ [p(x) \wedge \neg q(x)]$$

In words, the negation says, "There exists an integer x such that x is odd and $x^2 - 1$ is odd (that is, not even)." (This statement is false.)

Let $p(x, y)$, $q(x, y)$, and $r(x, y)$ represent three open statements, with replacements for the variables x, y chosen from some prescribed universe(s). What is the negation of the following statement?

$$\forall x\, \exists y\, [(p(x, y) \wedge q(x, y)) \rightarrow r(x, y)]$$

We find that

$$\begin{aligned}
& \neg[\forall x\, \exists y\, [(p(x, y) \wedge q(x, y)) \rightarrow r(x, y)]] \\
\Leftrightarrow\ & \exists x\, [\neg \exists y\, [(p(x, y) \wedge q(x, y)) \rightarrow r(x, y)]] \\
\Leftrightarrow\ & \exists x\, \forall y\, \neg[(p(x, y) \wedge q(x, y)) \rightarrow r(x, y)] \\
\Leftrightarrow\ & \exists x\, \forall y\, \neg[\neg[p(x, y) \wedge q(x, y)] \vee r(x, y)] \\
\Leftrightarrow\ & \exists x\, \forall y\, [\neg\neg[p(x, y) \wedge q(x, y)] \wedge \neg r(x, y)] \\
\Leftrightarrow\ & \exists x\, \forall y\, [(p(x, y) \wedge q(x, y)) \wedge \neg r(x, y)].
\end{aligned}$$